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SECTOR AT A GLANCE

QA and Standards

Skilling, Livelihoods and Labour Welfare Sector in India

DRAFT

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QA and Standards

Skilling Livelihoods and
Labour Welfare Sector in India

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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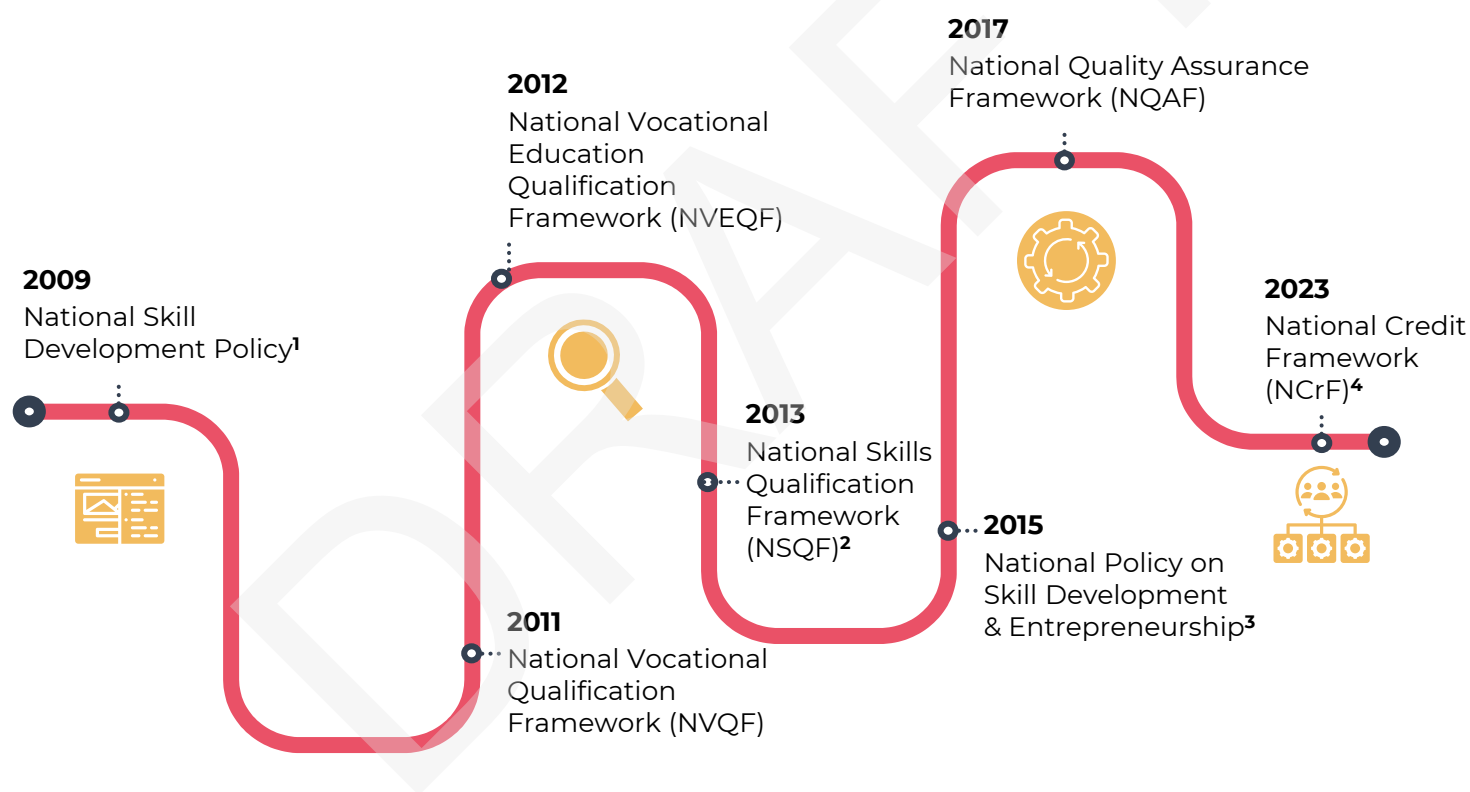
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Overview

The Government of India has recognized skill development as a major priority for India's socio-economic development, for sustaining its growth rate and the best means to convert India's demographics into an asset for development. The National Policy on Skill Development, 2009 brought focus on imparting training that is relevant to the market needs and covering diverse set of target groups. However, the initiative got a major push after the National Policy for Skill Development and Entrepreneurship 2015 was introduced. This policy focuses on **“Creating an ecosystem of empowerment by skilling on a large scale at speed with high standards and to promote a culture of innovation-based entrepreneurship to generate wealth and employment thus ensuring sustainable livelihoods for all citizens in the country”**.

The policy also paved the way for ensuring quality in training and laying out norms for standardizing the input for the skill training ecosystem. Standardization of processes for regulation and providing direction and flexibility to operating institutions to ensure quality outcomes in skilling initiatives, is essential to develop an effective and efficient skilling framework. Skill education and training institutions are governed by a quality-assurance framework that has evolved over the years.

Figure: 1 Evolution of Skilling Frameworks and Policies in India



The Ministry of Skill Development and Entrepreneurship (MSDE) set up the National Council for Vocational Education and Training (NCVET) in 2018 by merging National Skill Development Agency (NSDA) and National Council for Vocational Training (NCVT)⁵. NCVET was setup as a regulatory body for entities engaged in both short- and long-term skill training. It acts as a nodal agency to ensure quality assurance and standardization in the skilling ecosystem. NCVET primarily regulates awarding bodies, assessment agencies and assessors and trainers. MSDE, through National Skill Development Corporation (NSDC), also ensures quality of output by training providers through various measures.

¹Ministry Of Skill Development And Entrepreneurship. (2009). National Skill Development Policy.

²Ministry Of Education. (2013). National Skills Qualification Framework.

³Ministry Of Skill Development And Entrepreneurship. (2015). National Policy on Skill Development And Entrepreneurship.

⁴University Grants Commissions. (2023). National Credit Framework.

⁵National Council For Vocational Education And Training. (2018). About Us. Retrieved 6 July 2023

Sub-sectoral Trends

Qualification Standards and Awarding Bodies

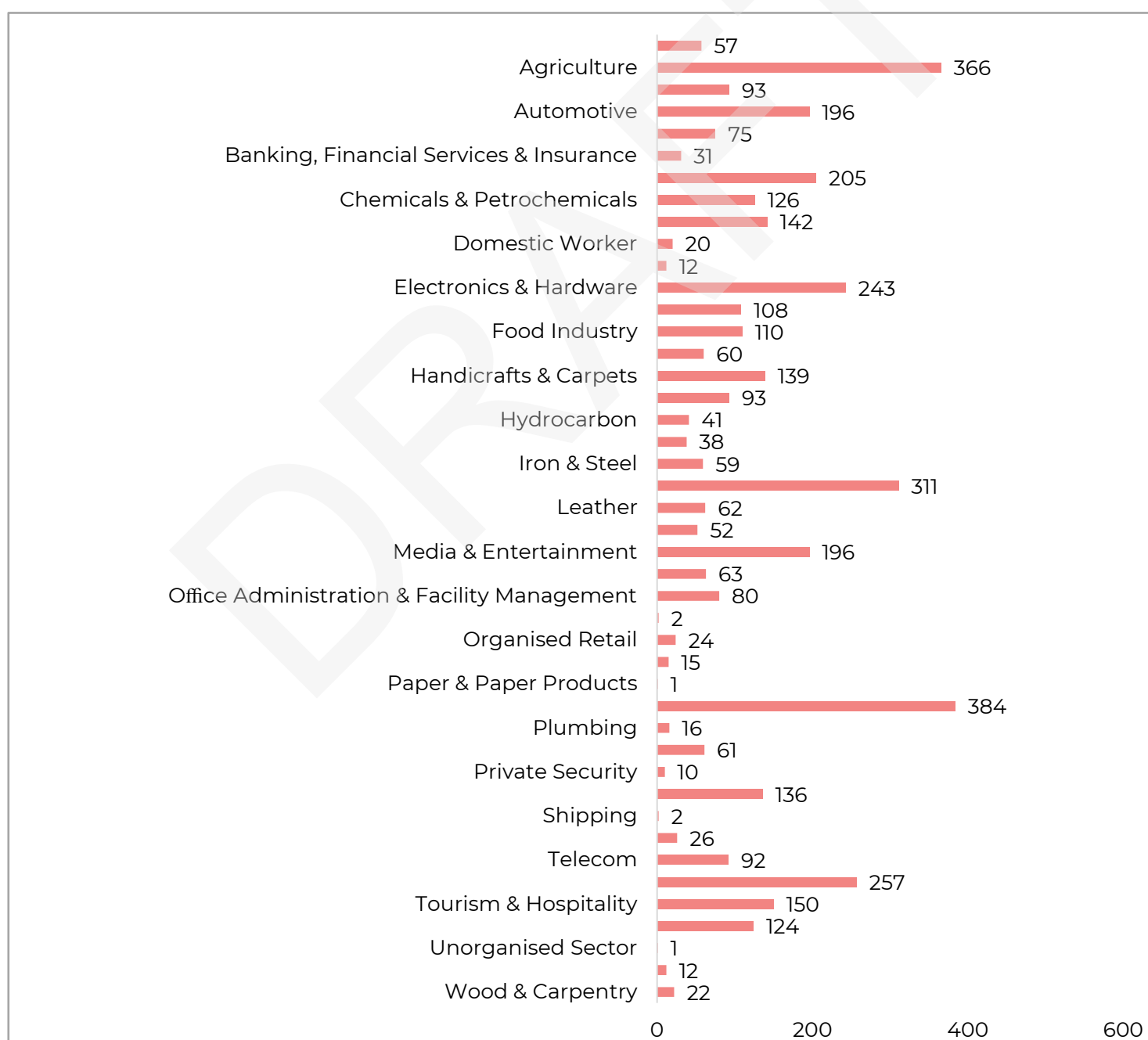
Qualification Standards

The qualification standards for vocational training are governed by the National Skills Qualification Framework (NSQF), which was notified on 27th December 2013. NSQF is a competency-based framework providing multiple pathways to facilitate mobility within vocational education and training (VET) and between VET and general education. NSQF has resulted in development of qualifications across sectors relevant to different learner groups.

The National Occupational Standards (NOS) describe the essential functions, standards of performance and knowledge/ understanding required for a job role. A Qualification Pack is a set of NOS, developed by the awarding body in alignment with the industry requirements, and updated from time to time.

80% of the standards are in 19 sectors highlighted in the dark colour in the graph

Figure 2: Sector-wise breakup of 4313 Qualification Standards across 57 Sectors



Source: National Qualification Register 1 (NQR) (Aug 2023)

National Council For Vocational Education And Training. (2018). National Qualification Register. Retrieved 6 July 2023

Types of Qualifications

Figure 3: Types of Qualifications



Source: National Qualification Register (Aug 2023)

National Credit Framework

The National Credit Framework (NCrF)⁶ has been jointly developed by the University Grants Commission, All India Council for Technical Education, National Council for Vocational Education and Training, National Institute of Open Schooling, Central Board of Secondary Education, National Council of Education Research and Training, Directorate General of Training, Ministry of Education, and Ministry of Skill Development and Entrepreneurship to achieve the vision and intent of National Education Policy'2020.

NCrF is a comprehensive single meta-framework document notified for creditisation and integrating learning in various dimensions of academics, skilling and experiential learning including relevant experience and proficiency/ professional levels acquired.

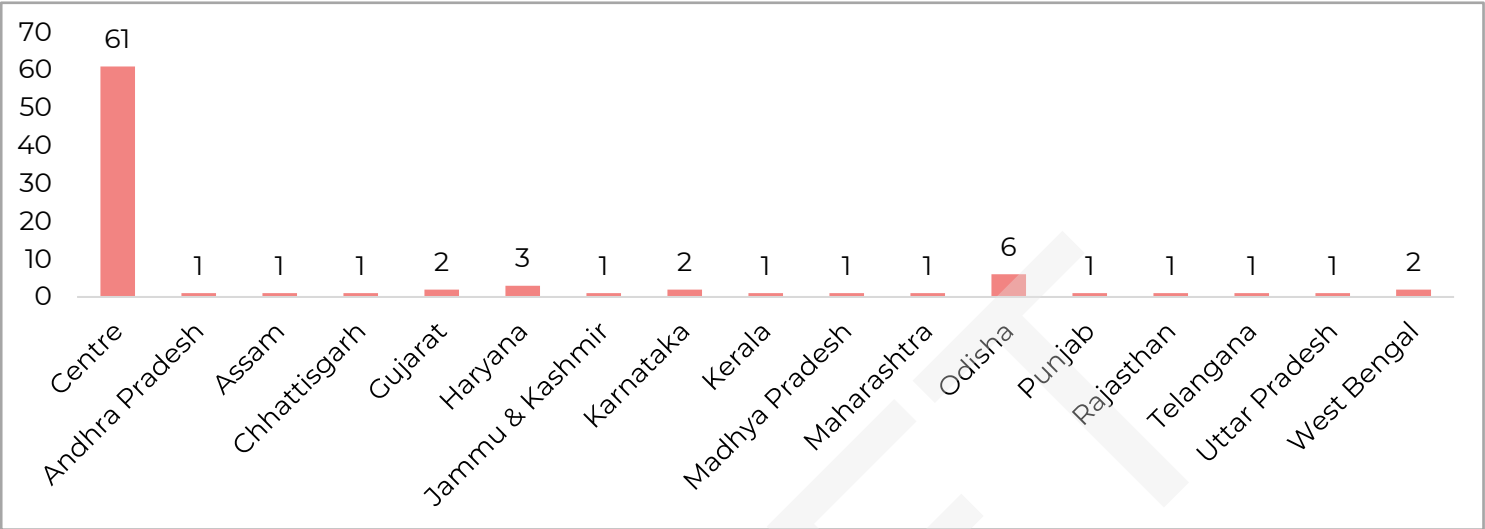
The National Credit Framework (NCrF) provides for the Assignment, Accumulation, Storage, Transfer and Redemption of Credits, and paves the way for multi-disciplinary education. It encourages the internationalization of education through credit transfer provisions, thus enabling wider recognition and acceptance of Indian education and skilling by other countries through international equivalence. The NCrF is operationalised through the Academic Bank of Credits (ABC); where students can choose their career paths, including an option for mid-way course corrections according to their talents and interests, and Multiple Entry-Multiple Exit (ME-ME) across the higher education, school education and vocational education.

⁶University Grants Commissions. (2023). *National Credit Framework*

Awarding Bodies

Any entity that awards or proposes to award certificates for a NSQF aligned and approved qualification is an Awarding Body⁷. It can provide the certificate for a developed qualification or an adopted one as per policy on adoption of Qualifications.

Figure 4: Number of Awarding Bodies Distributed by States



Source: National Council for Vocational Education and Training (Aug 2023)

Figure 5: Types of Awarding Bodies

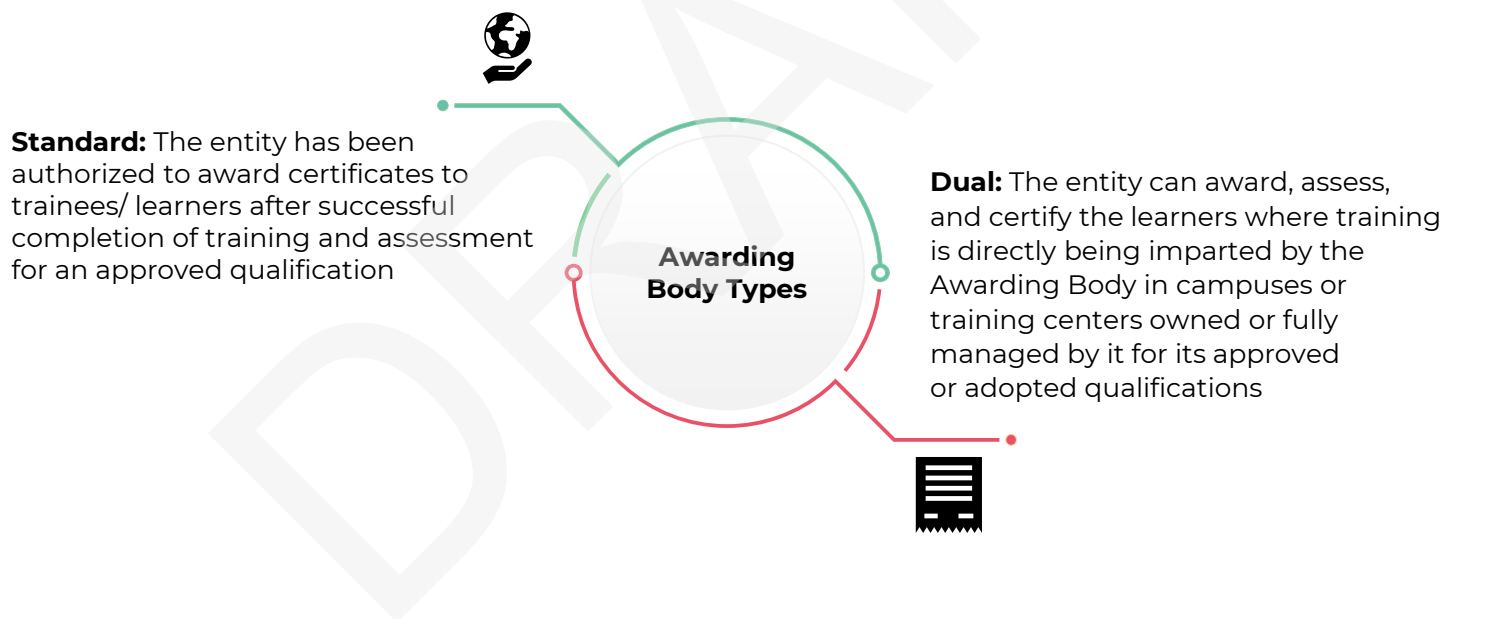
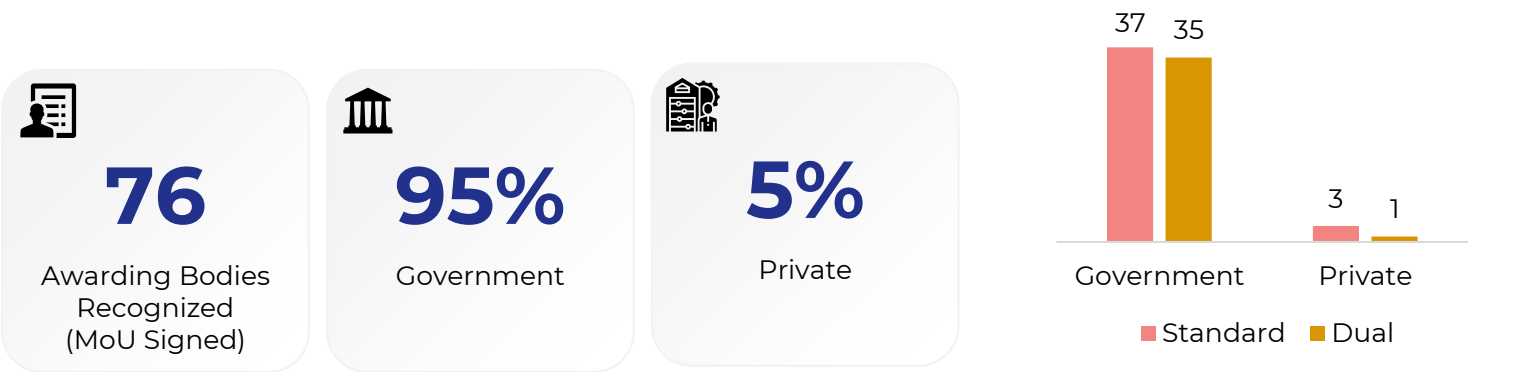


Figure 6: Number of Government and Private Awarding Bodies



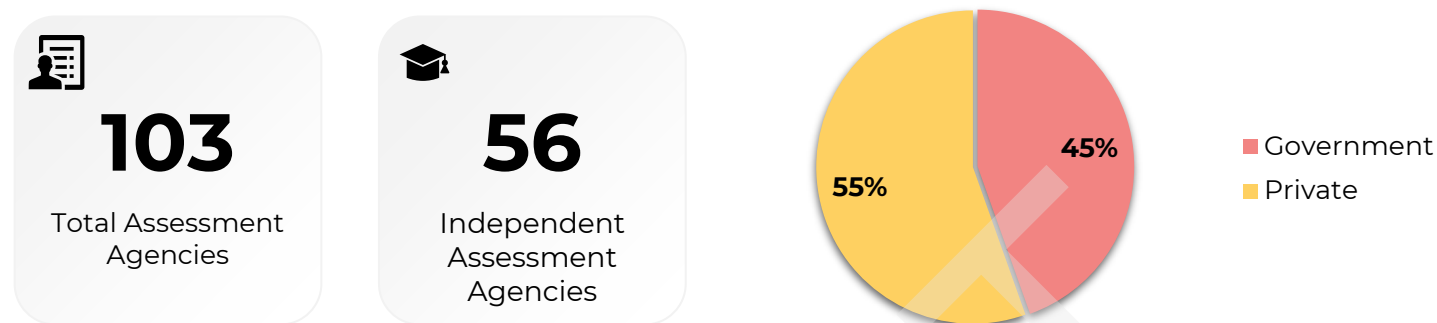
Source: National Council for Vocational Education and Training (Aug 2023)

⁷National Council For Vocational Education And Training. (2018). Recognition of Awarding Body. Retrieved 6 July 2023

Assessment Agencies

Assessment agencies⁸ tests or conducts examinations to assess whether a learner has met the requirements and is competent and qualified with respect to a skill or qualification. The regulation of assessment agencies is important to ensure that they have the required capacity and capability to conduct assessments in the designated sectors. In addition to the awarding bodies that have a dual recognition, there are independent organisations that are affiliated as assessment agencies.

Figure 7: Number of Government and Private Assessment Agencies



Source: National Council for Vocational Education and Training (Aug 2023)

Trainers and Assessors

Trainers and assessors play an important role in ensuring quality of training delivery and ensure that the desired learning outcomes are met. NCVET regulates the trainer and assessor ecosystem by laying out guidelines for their capacity building and recognition by the respective Awarding Bodies.

Figure 8: Total number of certified Trainers and Assessors

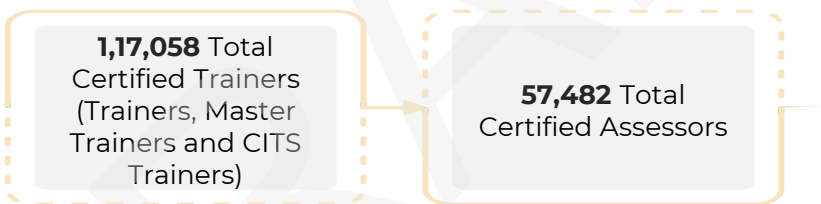
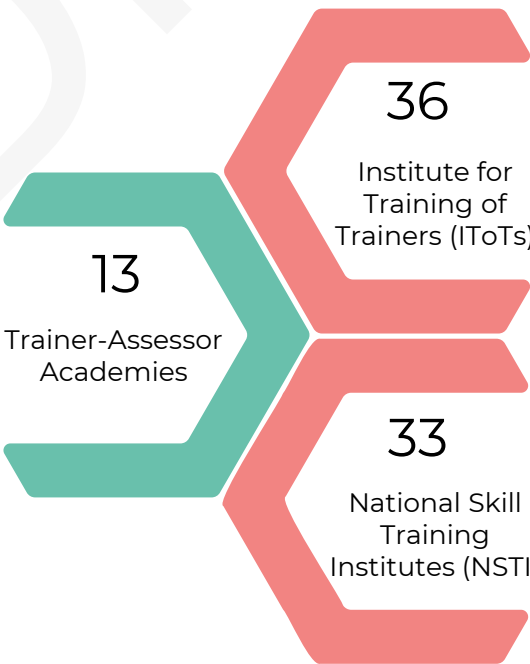


Figure 9: Types of Trainer Training Institutes in India

Trainers' Capacity Building - Key Initiatives

Centers of Excellences with strong industry linkages have been recognized as Trainer-Assessor Academies⁹ for training of trainers and assessors across 16 sectors. This initiative is anchored by NSDC and largely caters to Short term training ecosystem.



These are institutes set up under Directorate General of Training (DGT) to train the trainers of the Industrial Training Institutes (ITI). The institutes are attached with an ITI having the relevant trade for which training is provided.

These are premier institutes set up by Directorate General of Training (DGT) to impart training to the instructors of Industrial Training Institutes (ITI).

Source: NCVET, DGT, MSDE, NSDC (Aug 2023)

⁸National Council For Vocational Education And Training. (2018). Recognition of Assessment Agency. Retrieved 6 July 2023

⁹National Skill Development Corporation. (2008). Trainer-Assessor Academies. Retrieved 6 July 2023.

Trainers and Assessors Capacity Building Institutes Spread Across the Country

State	NSTI	NSTI and ITOT	NSTI, ITOT and Trainer Assessor Academy	NSTI and Trainer Assessor Academy	Trainer Assessor Academy
Andhra Pradesh	1	0	0	0	0
Assam	0	0	0	0	0
Bihar	0	0	0	0	0
Chhattisgarh	0	0	0	0	0
Goa	0	0	0	0	0
Gujarat	0	0	0	0	0
Haryana	0	0	0	0	0
Himachal Pradesh	0	0	0	0	0
Jammu and Kashmir	0	0	0	0	0
Karnataka	0	0	0	0	0
Kerala	0	0	0	0	0
Madhya Pradesh	0	0	0	0	0
Maharashtra	0	0	0	0	0
Manipur	0	0	0	0	0
Mizoram	0	0	0	0	0
Nagaland	0	0	0	0	0
Northeast	0	0	0	0	0
Odisha	0	0	0	0	0
Punjab	0	0	0	0	0
Rajasthan	0	0	0	0	0
Sikkim	0	0	0	0	0
Tamil Nadu	0	0	0	0	0
Telangana	0	0	0	0	0
Uttar Pradesh	0	0	0	0	0
West Bengal	0	0	0	0	0

Training Providers

The long-term training ecosystem has a set of pre-defined rules that govern the establishment of ITIs in the country. These rules are set by DGT or the State Technical Education Department and any private organisation willing to establish an ITI will require an approval of either of the agencies.

¹⁰Directorate General Of Training. (1945). Craft Instructor Training Scheme – List Of Institutes Offering CITS. Retrieved 6 July 2023

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Source: Data as on August 2022 (NSDC)

Opportunities And Challenges



OPPORTUNITIES

- Involving industry to actively participate in the development/ upgradation of required skills qualifications.
- Aligning standards to international norms to facilitate workforce mobility by harmonization of global skills standards.
- Agile mechanisms for capturing real-time demand and updating industry centric training standards as per the requirements.
- Formalization of vocational education by developing specific standards at school level under the ambit of National Education Policy 2020 to establish equivalence between general and vocational education and training.
- Creditization of learning for students with varying learning capacities and provisions for lifelong learning through “Multiple Entry and Multiple Exit” mode in alignment with the National Credit Framework, and utilising ‘Academic Bank of Credits’ for storing credits earned at different stages of learning.
- Enabling measures to make the skill training ecosystem responsive to the impact of technology and climate on jobs.
- Developing 21st century skills across domains to meet the emerging requirements of both manufacturing and services sector.



CHALLENGES

- Gap in industry requirements and skill standards due to a fast-evolving industry ecosystem.
- Inadequate number of certified Trainers and Assessors.
- Low awareness about NSQF aligned standards and certifications among employers.
- Developing employer-led demand for NSQF aligned certifications through a mix of demand and supply side incentives.
- Mapping all skill courses with National Classification of Occupations (NCO) which in term looks at global (ISCO) alignment.
- Standardization of training processes to ensure quality of skill training.

Cross Cutting Themes

Gender

Gender Sensitivity¹² has been developed to reduce barriers to personal and economic development created by sexism. There are five Gender Sensitivity (GS) levels that have been suggested by NSDC corresponding to the NSQF levels. They are flexible and can be adapted as per job role and sector. The GS Framework is structured on 3 parameters –

- Knowledge
- Behaviour and Practice
- Policies, Incentives and Penalties.

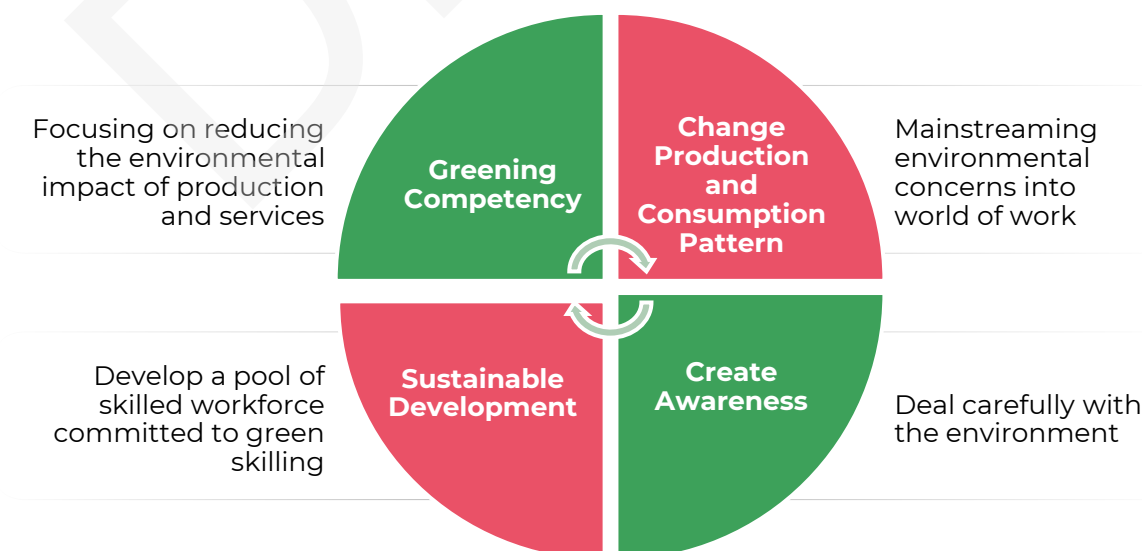
Suggested NSQF Level	Gender Sensitivity Level (GS)	Knowledge	Policies, Incentives and Penalties
9 and 10	5	Strategic/Extended Knowledge	Continuously innovating and improving on Practices
7 and 8	4	Structural/Procedural Knowledge	Standardizing practices and optimizing outcomes
5 and 6	3	Active Knowledge	Align processes and approaches
3 and 4	2	Causal Knowledge	Developed awareness and growing involvement
1 and 2	1	Foundational Knowledge	Growing awareness of laws and penalties

Source: NSDC

Climate

The Greening Framework¹³ seeks to include 'green knowledge' and 'green methods' in skill training. It enables greening of competencies in existing job profiles, with a focus on reducing the environmental impact of production, services and how people work

Figure 11: Greening Framework



¹²National Skill Development Corporation. (2008). Gender Sensitivity. Retrieved 6 July 2023.

¹³National Skill Development Corporation. (2008). Greening Framework. Retrieved 6 July 2023.

Proposed Greening Framework

Suggested Levels		Key Tasks		Implementation Level
	Solutions	Policies	Implementation	
8 to 10	Design greening solutions to make the organisation more sustainable	Design organisational greening policies in line with govt policies/acts of the sector to implement sustainability at workplace/ organisations	Strategize the implementation of greening policies at an organisational level	Management and Organisational Level
5 to 7	Interpret and implement greening solutions	Interpret and implement organisational greening policies and obtain required regulatory approvals	Supervise implementation of all greening practices Create awareness about the policies	Individual + Organisational Level
1 to 4	Follow greening solutions, policies and practices			Individual Level (Within or outside the organisation)

Future of Work

New age exponential technologies have been disrupting the process of the world of work, and this necessitates the need to address the existing skill gaps resulting from technological advancements, and advancing efforts focused on skilling, upskilling and re-skilling of individuals in the field of automation and artificial intelligence

5

Qualifications on Artificial Intelligence

2

Qualifications on AR/VR

15

Qualifications on Drone Technology

Different Types of Existing Future Skills Qualifications

9

Qualifications on Robotics

7

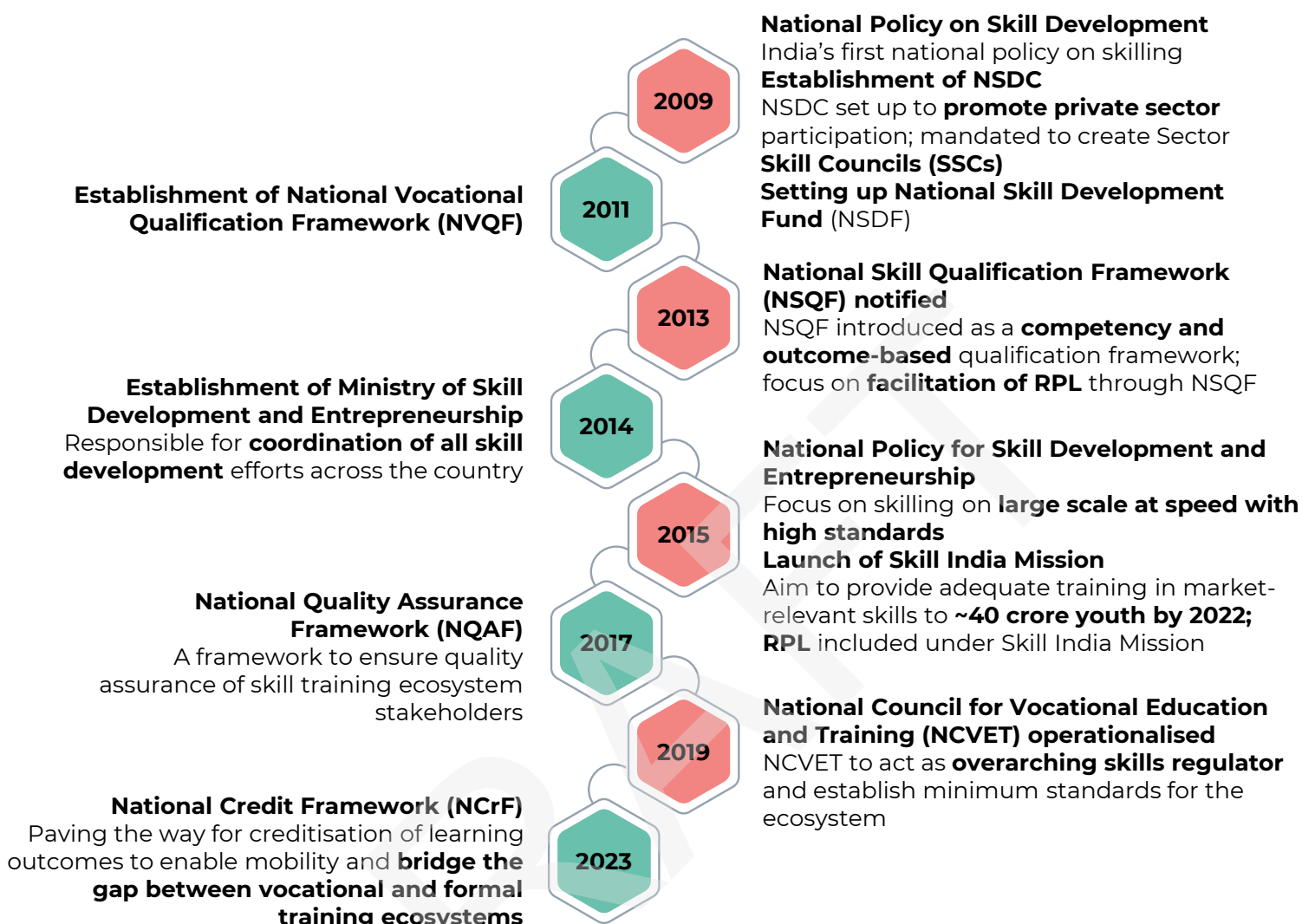
Qualifications on Internet of Things

9

Qualifications on Cloud Computing

Government Initiatives

Key Milestones



Government Initiatives

'**One Nation One Standard**' is the mantra to ensure that national standards and quality for skilling are globally aligned and Indian youth can aspire to secure local, national, and international job opportunities. Quality of training is measured by competency outcomes and employability of trainees and enabled by key Government institutions and initiatives.

Pivot Institutions

National Council for Vocational Education and Training (NCVET)	National Skill Development Corporation (NSDC)	Sector Skill Councils (SSCs)
NCVET acts as the overarching skills regulator and regulates the functioning of entities engaged in short-term training Main functions: - Recognise and regulate awarding bodies and assessment agencies - Approve qualifications developed by awarding bodies and SSCs - Monitor and supervise recognized entities - Grievance redressal	NSDC, set up as a public private partnership, has the mandate to catalyse the skilling landscape Main functions: - Implement Government schemes - Enable support systems such as quality assurance, and train the trainer academies - Develop appropriate models to enhance, support and coordinate private sector initiatives - Fund select private sector skilling initiatives - Provide knowledge services through skill gap studies, etc.	SSCs¹⁴ are set up as autonomous industry-led bodies by NSDC; 37 operational SSCs¹, with 600+ corporate representatives Main functions: - Create Occupational Standards and Qualification bodies - Develop competency framework - Conduct Train the Trainer Programs - Conduct skill gap studies - Assess and certify trainees on the curriculum aligned to National Occupational Standards developed by them

¹⁴National Skill Development Corporation. (2008). *Sector Skill Councils*. Retrieved 6 July 2023.

Key Policies, Schemes and Initiatives of the Government



National Skills Qualification Framework

Launched By: Government of India; and now under the ambit of “Ministry of Skill Development and Entrepreneurship”

Year of Launch: 2013

Objective: A Skilling Framework to organize qualifications according to a series of levels of knowledge, skills, and aptitudes. The levels are defined through learning outcomes.



National Quality Assurance Framework

Launched By: Ministry of Skill Development and Entrepreneurship¹⁵

Year of Launch: 2015

Objective: Aims to improve and provides the benchmarks or quality criteria which the different organisations involved in education and training must meet to be accredited to provide education and training/skills activities.



National Education Policy

Launched By: Ministry of Education¹⁶

Year of Launch: 2020

Objective: Based on the principle that education must develop not only cognitive capacities - both the ‘foundational capacities’ of literacy and numeracy and ‘higher-order’ cognitive capacities, such as critical thinking and problem solving – but also social, ethical, and emotional capacities and dispositions. Key elements of NEP is Awareness, Access, Accountability, Equity and Quality.



National Credit Framework

Launched By: Ministry of Skill Development and Entrepreneurship, Ministry of Education, and Ministry of Labour and Employment

Year of Launch: 2023

Objective: Single meta framework for creditisation and integrating learning in various dimensions of academics, skilling and experiential learning including relevant experience and proficiency/ professional levels acquired through schools, higher education institutions and vocational institutes.



Guidelines for Capacity Building

Launched By: Ministry of Skill Development and Entrepreneurship

Year of Launch: 2021

Objective: To train and create pool of trainers through various Government initiatives like NISHTHA, Craft Instructor Training Scheme (CITS), National Initiative for Training of Technical Teachers (NTTT) and Training of Trainers¹⁷ and Training of Assessors.

¹⁵Ministry Of Skill Development And Entrepreneurship. (2014). About Us. Retrieved 6 July 2023.

¹⁶Ministry Of Education.(1985). About Us. Retrieved 6 July 2023.

¹⁷National Council For Vocational Education And Training. (2018). Guidelines For Training Of Trainers. Retrieved 6 July 2023



Guidelines for Qualifications

Launched By: Ministry of Skill Development and Entrepreneurship

Objective: NCVET has laid guidelines for development of various types of National Occupational Standards and Qualifications¹⁸



Guidelines for Curricula and Content

Launched By: Ministry of Skill Development and Entrepreneurship

Objective: NSDC had laid guidelines for Model Curricula¹⁹ creation and skilling content creation



Assessment Guidelines²⁰

Launched By: Ministry of Skill Development and Entrepreneurship

Objective: To assess the competency of the candidate w.r.t the performance criteria of the Qualifications mapped with variables like time duration, allocation of scores, difficulty levels of items, the role of the assessor in the assessment, mode of assessment and language considerations etc.



Guidelines for Blended Learning²¹

Launched By: National Council for Vocational Education and Training (NCVET)

Objective: NCVET has laid guidelines for Blended Learning for Vocational Education, Training and Skilling to integrates technology and digital media-based learning with traditional instructor-led classroom teaching and learning, as well as blending it with hands on workshop and shop floor skills training/ learning activities, giving students more flexibility to customize the learning experiences.



Guidelines for Micro Credentials (MC)²²

Launched By: National Council for Vocational Education and Training (NCVET)

Objective: To create an enabling environment for Awarding Bodies to design and develop modular learning standards further empowering learners to have continuous learning and cross mobility after honing multiple skill sets. These Guidelines will bring into focus the high in-demand and emerging Industry skills thereby providing larger avenues for growth and enhancement of skills through flexible modules of learning.

¹⁸National Council For Vocational Education And Training. (2018). Qualification Related. Retrieved 6 July 2023

¹⁹National Skill Development Corporation. (2008). Curriculum Template. Retrieved 6 July 2023

²⁰National Skill Development Corporation. (2008). Assessment Process And Protocols. Retrieved 6 July 2023

²¹National Council For Vocational Education And Training. (2018). Guidelines For Blended Learning, Education And Skilling. Retrieved 6 July 2023

²²National Council For Vocational Education And Training. (2018). Guidelines For Micro Credentials. Retrieved 6 July 2023

Way Forward

Relevant stakeholders can help to improve the quality of skill training programmes by improvising standards for the quality of vocational education and training, establishing performance assessments and evaluation mechanisms, and disseminating information about the providers.



Improving relevance of skill standards

- Industry adoption of skill standards to ensure that all type of skill training is recognized for employment by the industry.
- Alignment of skill standards to international standards, including global, regional and country specific standards. This will enable worker mobility and provide international recognition to skill training in India. The India-Saudi skill harmonization framework is one such example.
- Creditisation of all courses to ensure that academic credits are provided for skill training and enabling vertical and horizontal mobility.
- Enabling micro-credential-based training programs to promote multiskilling and providing flexibility to academia to develop industry specific modules. The initiative has been taken up under NCVET and has seen participation of several Sector Skill Councils like SSC NASSCOM.
- Focus on anticipating skill shifts in the future and developing standards to cater to new skill requirements.
- Promoting industry led institutions as awarding bodies. Industry led institutions provide training as per current industry requirements and their curriculum can be adopted as standards to improve relevance of skill training.



Strengthening skill training ecosystem stakeholders

- To impart latest industrial knowledge to training facilitators. Updating training infrastructure and learning resources requirements to include the latest technology and tools being used in the industry.
- Promoting mentorship and incentivizing industry practitioners to take up skill training to develop an ecosystem that can foster work-integrated learning.
- Capacity building of teaching and training staff on latest pedagogical/ andragogical techniques to improve training effectiveness. States like Madhya Pradesh have collaborated with leading institutes like IIM to build capacity of their TVET instructors and staff.
- Expanding industry led trainer-assessor academies. Industry driven institutes like SDI Bhubaneswar have been recognized as trainer-assessor academies.
- Promoting use of technology in training for standardization of processes.
- Promoting industry led training organisations and providing incentives to industry to enter skill training. Industries like Daikin, Lumax Technologies, Maruti are extensively partnering with Government institutions to impart industry relevant skill training.

Recommended Readings

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